

Sherwin Chiu

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Skills

Languages C++, C, Rust, Typescript, Python, Java, Objective C, ARM/RISC-V, SQL, MATLAB, HTML
Software GCP, AWS, .NET, Jenkins, Git, Docker, Kubernetes, Terraform, Bitbucket, Jira, Confluence

Education

University of Waterloo – Honours Bachelor of Applied Science, Computer Engineering, Co-op 2020 – 2025 Waterloo, ON
Relevant Courses: Distributed Computing, Real-Time Operating Systems, Compilers, Concurrency

Experience

Software Engineer Intern

January 2024 – April 2024

Geotab Inc., Waterloo, Ontario

- Deployed automated **CI/CD pipelines** using GitLab pipelines to streamline builds, testing, and deployments
- Created specific Mappers to fetch data from Geotab API to route to Geotab Admin through **Pub/Sub**
- Implemented **Terraform IaC** to automate cloud resources and built a **Webhook** endpoint for Hyundai OEM
- Migrated legacy code from **Virtual Machines to Kubernetes** to increase throughput and code readability

Software Engineer Intern

January 2023 – April 2023

Motive Technologies Inc., San Francisco, California

- Integrated code analysis tools within **Jenkins pipelines** to increase code coverage from **25% to 34%**
- Resolved critical **concurrency** and **multithreading** issues in the Motive Mobile Platform codebase, improving background service reliability and offline data synchronization by reducing mobile crashes

Software Engineer Intern

May 2022 – October 2022

Swift Medical Inc., Toronto, Ontario

- Constructed **iOS** application to capture images of chronic wounds and process hyperspectral images
- Developed contour algorithms in **OpenCV2** to detect bacteria and hemoglobin levels in chronic wounds
- Implemented Bluetooth commands between hyperspectral device used for medical imaging and iOS app

Projects

Electrium Mobility Founder 🔗

November 2022 – January 2025

University of Waterloo Design Team, Waterloo, Ontario

- Founded a student **design team** promoting sustainable cities and mentoring **500+** engineering students
- Designed and built custom electric vehicles, developing **ESCs, precharge circuits, and BLE remote controls**
- Led all major sub teams, overseeing **hardware, firmware and software** development and producing 8+ EVs
- Maintained a website with over **500+** visits monthly which uses Cloudflare and Github Pages

ElectriCITY 🔗

Bigmode Game Jam 2025

- **Developed a strategy-based video game** where players build transmission towers to connect energy producers and consumers, managing an evolving energy grid using GameMaker as a game engine
- **Designed resource management mechanics** balancing energy production and consumption

Extracurricular & Awards

Hardware Volunteer Lead, Hack the North

June 2023 – September 2024

Vice President of Outgoing Exchange, AIESEC at Waterloo

January 2022 – December 2023

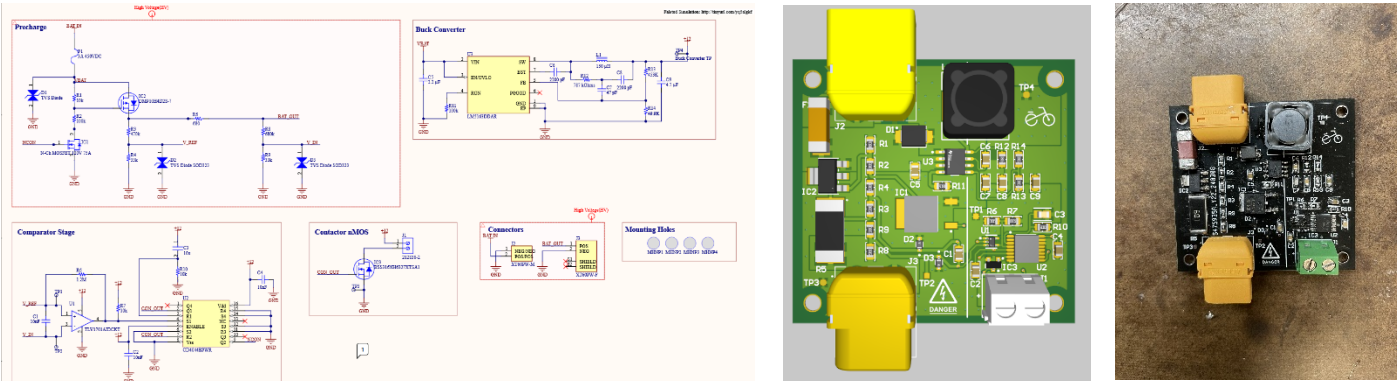
Schulich Leader Nominee, Schulich Leader Scholarships

January 2020 – June 2020

Project Portfolio

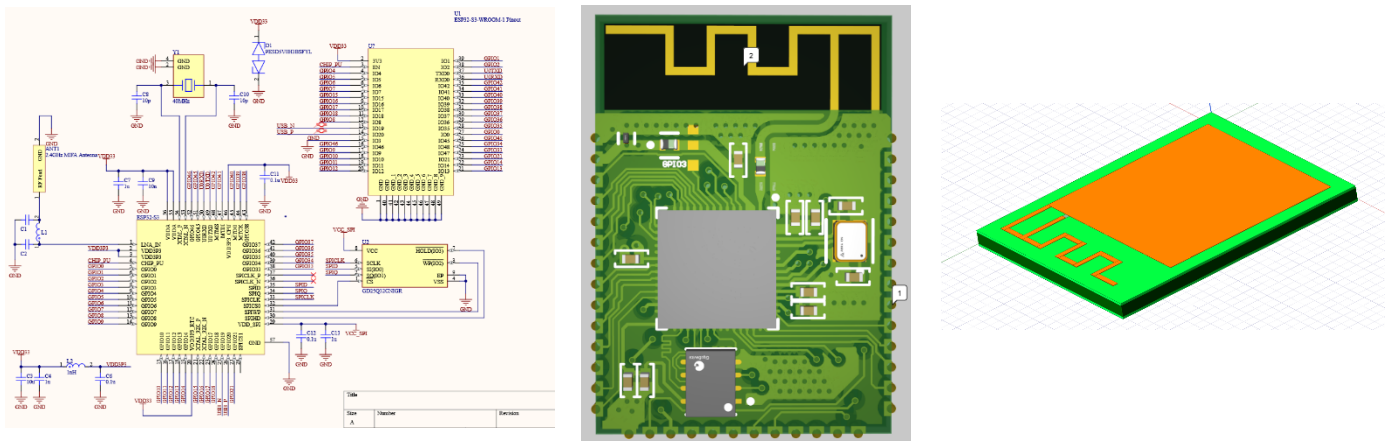
Electrium Pre-charge for Electric Motorcycle

- Created pre-charge circuitry to limit in-rush current entering motor controller which is caused by initial connection of a capacitive load to a battery, as the capacitive load acts as a “short circuit”
- Designed and simulated schematic using a TLV1701AIDCKT comparator to monitor capacitive load charge
- Used an initial charge resistor of 680 ohms to reach 99.328% of the voltage (5 tau of capacitive load), then switch fully to using a battery using a contactor after capacitor is detected as fully charge



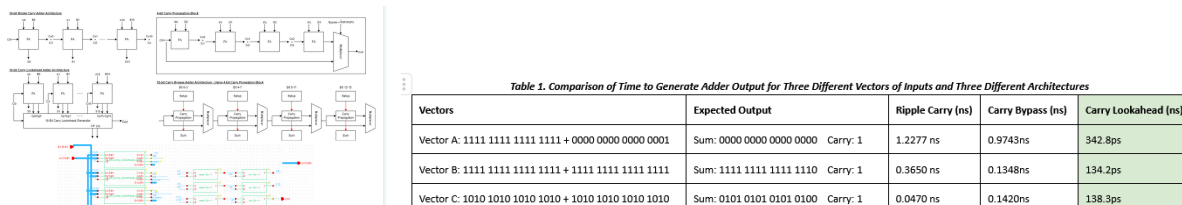
ELECTREX-32 Custom ESP32 Dev-Board

- Created custom ESP32 Dev Board using a ESP32-S3 IC and utilizing castellated holes on the PCB
- Simulated antenna design to achieve desired results for WiFi and BLE transmission



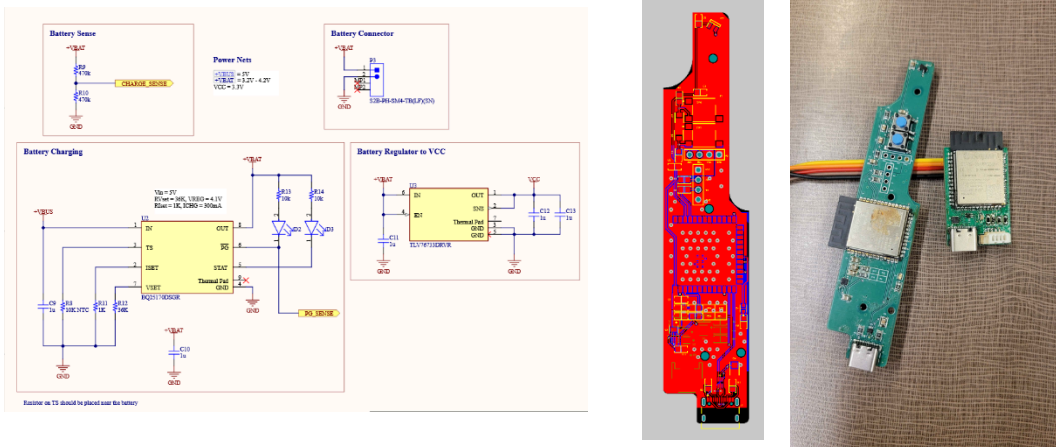
16-Bit Adder Design

- Utilized Cadence Virtuoso software to build multiple architectures of 16-bit adders and scaling PMOS and NMOS widths based on propagation delays such as carry-bypass, carry-lookahead, and ripple carry
- Simulated propagation delays of different adder architectures in different temperature corner situations



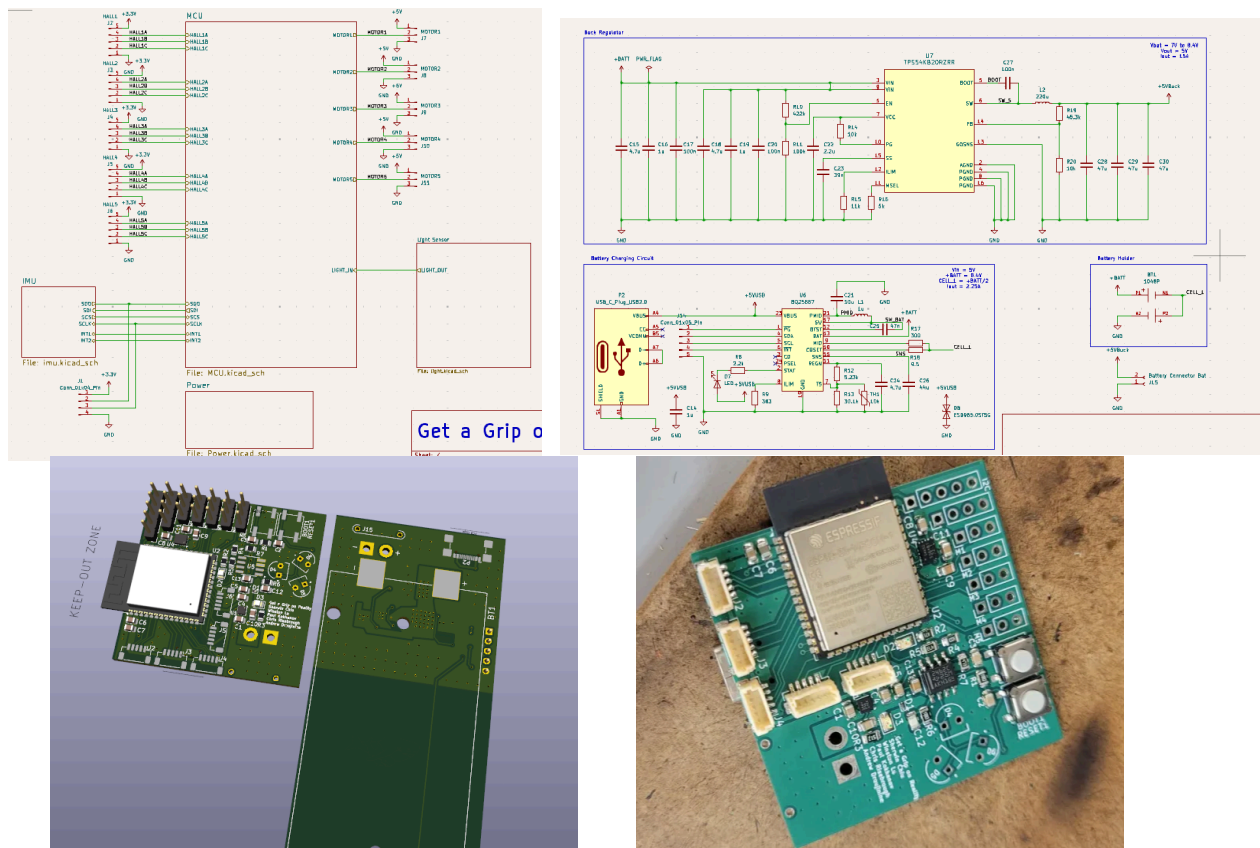
Electrium Remote Control

- Designed a custom remote-control PCB to interface with motor controllers using an ESP32
- Use BQ25170DSGR IC for Lithium-Polymer battery charging and a TLV76733DRVR linear regulator for 3.3V
- Make use of rotating magnets and a hall effect sensor for accurate analog reading of signals



Final Year Design Project, Capstone – Get a Grip on Reality

- Capstone focused on making force-feedback virtual reality gloves by integrated physical gloves that apply pressure with virtual reality programs for realistic hand feedback when grabbing objects
- Designed custom PCB designed around an ESP32-S3 to control servo motors, hall effect sensors, IR light sensors, and an IMU for tracking position and to provide force on each hand
- Designed custom power distribution board which outputs 5V @ up to 10A using TPS54KB20RZRR step down converter, charged using a BQ25887 boost battery charger and powered by 2 Lithium-Ion Batteries



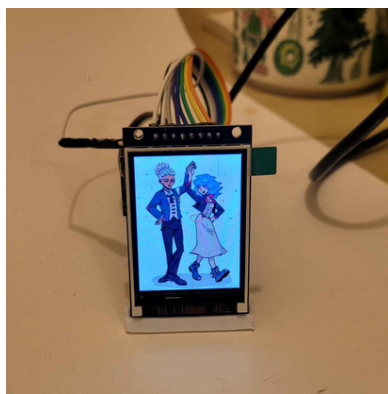
Arcane Clap Light – YouTube: [Link Here](#)

- In Arcane, there's a scene in the mines where Vi and Jinx go into the mines, and I replicated the little blue glowing mushrooms that light up anytime a loud sound is detected such as a clap or slam
- Used an ESP32-C3 Super Mini as the microcontroller, a sound sensor, a MAX98357 I2S DAC with speaker and 5mm LEDs as the light source controlled with PWM and BJTs to dial in the color
- Sounds clips were converted to 16-bit PCM and stored in the flash memory of the ESP32-C3 (4MB of flash!)
- Quickly soldered and prototyped on protoboard for a prototype concept with 3D Printed Stand



Wireless Display for Images

- I created a wireless display (240x320) which fetches images from a GCP server and displays it
- Utilized Google Cloud Run and Google Bucket for server to store images, and Netlify for client
- Made with an ESP32-C3 which uses WiFi to connect to the server and an ST7789 2" LCD for images
- Programmed firmware in Arduino which fetches an image, unpackages bitmap to RGB565 to display



Pixplay

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Below View is Image Shown on Display

